

Arpit Garg

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Summary

AI/ML Engineer with experience in machine learning, deep learning, computer vision, and Spiking Neural Networks. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, and OpenCV, with expertise in model training, evaluation, and feature engineering.

Education

University of Delhi <i>B.Tech in Electronics and Communication Engineering (Minor in AI/ML)</i> CGPA: 7.68/10 (up to 6th semester)	Aug 2023 – Present Delhi, India
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Relevant Coursework

Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Neural Networks, Deep Learning, Natural Language Processing, Digital Signal Processing, Probability & Statistics, Reinforcement Learning, Data Analytics, Edge Computing.

Experience

Research Intern(Remote) – IIT Roorkee <i>Machine Learning Research</i>	May 2026 – Jul 2026 Roorkee, India
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- Studied SNN architectures, surrogate-gradient learning, spike-based computation, and 3 neuromorphic learning methodologies.
- Designed a multi-class classification framework achieving approximately **70% accuracy** using separate hard and soft classifiers selected based on prediction confidence.

Summer Intern (On-Site) – Warship Design Bureau, Indian Navy <i>Project Management & Systems Coordination</i>	Jun 2026 – Jul 2026 New Delhi, India
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- Studied project planning, resource coordination, and workflow processes across 5+ indigenous warship development programs.
- Coordinated with 6 multidisciplinary teams across electrical, weapon, navigation, communication, and underwater systems during project execution and system integration.
- Analyzed project dependencies, integration of 100+ equipment units, and cross-functional processes involved in naval engineering projects.

Research Intern – IIIT Delhi <i>Department of Computational Biology</i>	Jun 2025 – Aug 2025 Delhi, India
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- Constructed predictive models using ECFP6 and GROVER molecular fingerprints, improving performance through feature engineering and hyperparameter optimization.
- Analyzed over **10,000** protein–protein interaction samples and validated results using k-fold cross-validation and PR-curve benchmarking.

Projects

Handwriting Synthesis using RNNs 🧠	Jan 2026 – Feb 2026
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- * Trained an LSTM-based sequence-to-sequence model to synthesize handwritten strokes with preprocessing and visualization pipelines, achieving **82% accuracy**.
- * Applied temporal sequence modeling to generate realistic handwriting and evaluated performance across **500+ validation sequences**.

Image Captioning using Deep Learning 🧠	Nov 2025 – Dec 2025
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- * Engineered a CNN-LSTM image captioning pipeline trained on approximately **8,000 images**, combining convolutional feature extraction with recurrent sequence generation.
- * Established a **5-stage** preprocessing and caption generation workflow including image preprocessing, tokenization, sequence padding, model training, and inference using TensorFlow/Keras.
- * Achieved a **BLEU-4 score of 0.34** through model training, preprocessing, and hyperparameter tuning.

FitBudd – AI-based Fitness Assistant

Jun 2025 – Aug 2025

- * Created a fitness analysis application using MediaPipe Pose with **33 body keypoints** for real-time posture estimation and exercise analysis.
- * Devised skeletal feature extraction using joint-angle computation, motion velocity estimation, and temporal smoothing, reducing pose estimation noise by approximately **20%**.
- * Integrated pose estimation, exercise classification, and posture feedback into a workout monitoring framework supporting 18+ exercise types.

Technical Skills

Programming: Python, C++, C, JavaScript, SQL, HTML, CSS

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Hyperparameter Optimization, Cross-Validation, Ensemble Learning

Deep Learning: ANN, CNN, RNN, LSTM, SNN, GNN, Sequence-to-Sequence Models, Transfer Learning

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, SpikingJelly

Data Science: NumPy, Pandas, SciPy, Matplotlib, Data Preprocessing, Exploratory Data Analysis

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Docker, Linux

Achievements

2nd Place – Code or Die Hackathon, Delhi Technological University, for implementing a software solution under hackathon constraints.

Smart India Hackathon 2025 – Shortlisted (Waitlist): Selected among 600+ shortlisted national teams for the PeerFit project based on its technical proposal and proposed solution.

Leadership & Positions of Responsibility

- **Finance Head – Training and Placement Cell** Aug 2026 – Present
Handled a budget exceeding **INR 60 lakh**, including expense tracking, fund allocation, and financial planning for placement-related activities and recruitment operations.
- **Joint Treasurer – IEEE Student Branch** Jan 2026 – Present
Oversaw budgeting, fund allocation, and expense tracking for 6+ student technical activities and events.
- **Volunteer – NVIDIA Community Meetup** October 2025
Supported event coordination, participant registration, logistics, and technical event operations for a community meetup with over **1,500 attendees**.